

Names	Solar Probe (before 2002) Solar Probe Plus (2010–2017) Parker Solar Probe (since 2017)
Mission type	<u>Heliophysics</u>
Operator	<u>NASA / Applied Physics Laboratory</u>
COSPAR ID	2018-065A (https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2018-065A)
SATCAT no.	43592 
Website	<u>parkersolarprobe.jhuapl.edu</u> (http://parkersolarprobe.jhuapl.edu)
Mission duration	7 years (planned) Elapsed: 7 years, 9 months and 14 days

Manufacturer	<u>Applied Physics Laboratory</u>
Launch mass	685 kg (1,510 lb) ^[1]
Dry mass	555 kg (1,224 lb)
Payload mass	50 kg (110 lb)
Dimensions	1 × 3 × 2.3 m (3.3 × 9.8 × 7.5 ft)
Power	343 W (at closest approach)

Launch date	12 August 2018, 07:31 UTC ^{[2][3]}
Rocket	<u>Delta IV Heavy / Star 48BV</u> ^[4]
Launch site	<u>Cape Canaveral, SLC-37</u>
Contractor	United Launch Alliance

Reference system	<u>Heliocentric orbit</u>
Semi-major axis	0.388 AU (58.0 million km; 36.1 million mi)
Perihelion altitude	0.046 AU (6.9 million km; 4.3 million mi; 9.86 <i>R</i> _⊙) ^[note 1]
Aphelion altitude	0.73 AU (109 million km; 68 million mi) ^[5]
Inclination	3.4°
Period	88 days

Band	<u>K_a-band</u> , <u>X-band</u>
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Large Strategic Science Missions

***Living With a Star* program**

It used repeated gravity assists from Venus to develop an eccentric orbit, approaching within 9.86 solar radii (6.9 million km or 4.3 million miles)^{[7][8]} from the center of the Sun. At its closest approach in 2024, its speed relative to the Sun was 690,000 km/h (430,000 mph) or 191 km/s (118.7 mi/s), which is 0.064% the speed of light.^{[7][9]} It is the fastest object ever built on Earth.^[10]

On 29 October 2018, at about 18:04 UTC, the spacecraft became the closest ever artificial object to the Sun. The previous record, 42.73 million kilometers (26.55 million miles) from the Sun's surface, was set by the *Helios 2* spacecraft in April 1976.^[13] At its perihelion on 27 September 2023, the PSP's closest approach was 7.26 million kilometers (4.51 million miles),^[14] reaching this distance again on 29 March 2024.^[15]

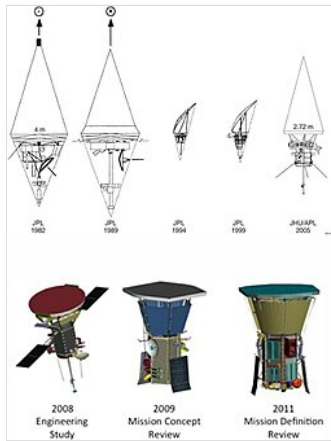
On 24 December 2024 at 11:53 UTC, PSP made its closest approach to the Sun, coming to a distance of 6.1 million km (3.8 million miles) from the surface. Its beacon signal was received on 26 December, showing that it had survived the passage through the corona.^{[16][17]} Detailed telemetry was received 1 January 2025.^[18]

In 2025, the teams from NASA, Johns Hopkins, and partners were awarded the 2024 Collier Trophy for their achievements.^[19]

History

The Parker Solar Probe concept originates in the 1958 report by the Fields and Particles Group, Committee 8 of the National Academy of Sciences' Space Science Board,^{[20][21][22]} which proposed several space missions including "a solar probe to pass inside the orbit of Mercury to study the particles and fields in the vicinity of the Sun".^{[23][24]}

Studies in the 1970s and 1980s reaffirmed its importance,^[23] but it was always postponed due to cost. A cost-reduced Solar Orbiter mission was studied in the 1990s, and a more capable Solar Probe mission served as one of the centerpieces of the Outer Planet/Solar Probe (OPSP) program formulated by NASA in the late 1990s. The first three missions of the program were planned to be: the *Solar Orbiter*, the Pluto and Kuiper belt reconnaissance Pluto Kuiper Express mission, and the Europa Orbiter astrobiology mission focused on Europa.^{[25][26]}



Evolution of PSP design

The original Solar Probe design used a gravity assist from Jupiter to enter a polar orbit which dropped almost directly toward the Sun. While this explored the important solar poles and came even closer to the surface ($3 R_{\odot}$, a perihelion of $4 R_{\odot}$),^[26] the extreme variation in solar irradiance made for an expensive mission and required a radioisotope thermal generator for power. The trip to Jupiter also made for a long mission, $3\frac{1}{2}$ years to first solar perihelion, 8 years to second.

Following the appointment of Sean O'Keefe as Administrator of NASA, the entirety of the OPSP program was canceled as part of President George W. Bush's request for the 2003 United States federal budget.^[27] Administrator O'Keefe cited a need for a restructuring of NASA and its projects, falling in line with the Bush Administration's wish for NASA to refocus on "research and development, and addressing management shortcomings".^[27]

In the early 2010s, plans for the Solar Probe mission were incorporated into a lower-cost *Solar Probe Plus*.^[28] The redesigned mission uses multiple Venus gravity assists for a more direct flight path, which can be powered by solar panels. It has a higher perihelion, reducing the demands on the thermal protection system.

In May 2017, the spacecraft was renamed the Parker Solar Probe in honor of astrophysicist Eugene Newman Parker,^{[29][30]} who had proposed the existence of nanoflares as an explanation of coronal heating^[31] as well as having developed a mathematical theory that predicted the existence of solar wind.^[32] The solar probe cost NASA US\$1.5 billion.^{[33][34]} The launch rocket bore a dedication in memory of APL engineer Andrew A. Dantzler who had worked on the project.^[35]

A memory card containing names submitted by over 1.1 million people was mounted on a plaque and installed below the spacecraft's high-gain antenna.^[36] The card also contains photos of Parker and a copy of his 1958 scientific paper predicting important aspects of solar physics.^[37]

Spacecraft

The Parker Solar Probe is the first spacecraft to fly into the low solar corona. It will assess the structure and dynamics of the Sun's coronal plasma and magnetic field, the energy flow that heats the solar corona and impels the solar wind, and the mechanisms that accelerate energetic particles.

The spacecraft's systems are protected from the extreme heat and radiation near the Sun by a solar shield. Incident solar radiation at perihelion is approximately 650 kW/m^2 , or 475 times the intensity at Earth orbit.^{[1][38]:31} The solar shield is hexagonal, mounted on the Sun-facing side of the spacecraft, 2.3 m (7 ft 7 in) in diameter,^[39] 11.4 cm (4.5 in) thick, and is made of two panels of reinforced carbon-carbon composite with a lightweight 11-centimeter-thick (4.5 in) carbon foam core,^[40] which is designed to withstand temperatures outside the spacecraft of about $1,370 \text{ }^{\circ}\text{C}$ ($2,500 \text{ }^{\circ}\text{F}$).^[1] The shield weighs only 73 kilograms (160 lb) and keeps the spacecraft's instruments at $29 \text{ }^{\circ}\text{C}$ ($85 \text{ }^{\circ}\text{F}$).^[40]

A white reflective alumina surface layer minimizes absorption. The spacecraft systems and scientific instruments are located in the central portion of the shield's shadow, where direct radiation from the Sun is fully blocked. If the shield was not between the spacecraft and the Sun, the probe would be damaged and become inoperative within tens of seconds. As radio communication with Earth takes about eight minutes in each direction, the Parker Solar Probe has to act autonomously and rapidly to protect itself. This is done using four light sensors to detect the first traces of direct sunlight coming from the shield limits and engaging movements from reaction wheels to reposition the spacecraft within the shadow again. According to project scientist Nicky Fox, the team described it as "the most autonomous spacecraft that has ever flown".^[6]

The primary power for the mission is a dual system of solar panels (photovoltaic arrays). A primary photovoltaic array, used for the portion of the mission outside 0.25 au, is retracted behind the shadow shield during the close approach to the Sun, and a much smaller secondary array powers the spacecraft through closest approach. This secondary array uses pumped-fluid cooling to maintain operating temperature of the solar panels and instrumentation.^{[41][42]}



A light bar testing in the Astrotech processing facility.



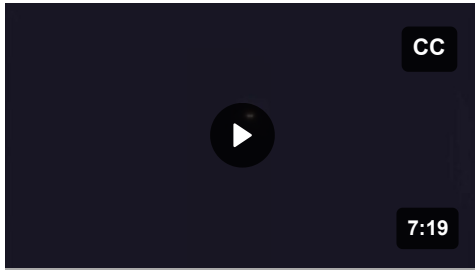
Parker Solar Probe mated to its third stage rocket motor



PSP during extensive environmental testing.



PSP encapsulated in fairing.



The launch of the probe.

Trajectory

The Parker Solar Probe mission design used repeated gravity assists at Venus to incrementally decrease its orbital perihelion to achieve a final altitude (above the surface) of approximately 8.5 solar radii, or about 6×10^6 km (3.7×10^6 mi; 0.040 au).^[39] The spacecraft trajectory included seven Venus flybys over nearly seven years to gradually shrink its elliptical orbit around the Sun, for a total of 24 orbits.^[1] The near Sun radiation environment was predicted to cause spacecraft charging effects, radiation damage in materials and electronics, and communication interruptions, so the orbit is highly elliptical with short times spent near the Sun.^[38]

The trajectory required high launch energy, so the probe was launched on a Delta IV Heavy launch vehicle and an upper stage based on the Star 48BV solid rocket motor.^[38] Interplanetary gravity assists provided further deceleration relative to its heliocentric orbit, which resulted in a heliocentric speed record at perihelion.^{[41][43]} As the probe passed around the Sun in December 2024, it achieved a velocity of 690,000 km/h (430,000 mph) or 191 km/s (118.7 mi/s) in the heliocentric ecliptic reference frame, which temporarily made it the fastest human-made object, almost three times as fast as the previous record holder, Helios-2.^{[7][9][44][45][46]}

Launch injection was very close to predictions, but nevertheless required path correction. Trajectory was re-optimized after the launch to save fuel. The first Venus flyby was only 52 days after the launch; three trajectory correction maneuvers were performed in this window.^[22]

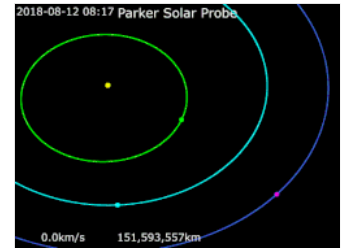
As described by Kepler's laws of planetary motion, which apply to any object in an orbit, gravity will cause the spacecraft to accelerate as it nears perihelion, then slow down again afterward until it reaches its aphelion. Because of its highly elliptical orbit and the Sun's strong gravity, this effect is particularly pronounced for the Parker Solar Probe. During a perihelion on September 27, 2023, the spacecraft traveled at 394,736 miles per hour (176.5 km/s), fast enough to fly from New York to Tokyo in just over a minute.^[14]

Final orbit, possible extensions and end of mission

The PSP performed its final gravity assist on November 6, 2024. It set the spacecraft on a new orbit passing 6.1 million kilometers (3.8 million miles) from the surface of the Sun.^[47] A beacon transmission was made and received successfully on December 20 to confirm that the craft was operating normally ahead of the perihelion. The exact time of closest approach was 11:53 UTC on December 24 but the craft was out of contact at this time. A further beacon transmission confirming successful passage was received on December 26.^{[48][49]}

This final orbit is inside the orbit of Venus and so no further encounters with that planet are planned. PSP will continue in this orbit but requiring adjustment to maintain attitude so that its transmitters point at Earth. Eventually its thrusters will run out of fuel and full functioning will not then be possible. The plan is to then rotate the craft so that its instruments will be exposed to the full radiance of the Sun for the first time. This is expected to ablate and destroy them. The heat shield will remain and is expected to continue to orbit the Sun for millions of years.^[50]

PSP's mission can be extended beyond main mission that will end in 2025. Currently discussed plan is "to complete the solar cycle at the 9.86 Rs perihelion distance".^[51]



An animation of the Parker Solar Probe's trajectory from August 7, 2018, to August 29, 2025:

Parker Solar Probe · Sun · Mercury · Venus · Earth
For more detailed animation, see [this video](#).



An apparent size of the Sun as seen from the Parker Solar Probe at perihelion compared to its apparent size seen from Earth

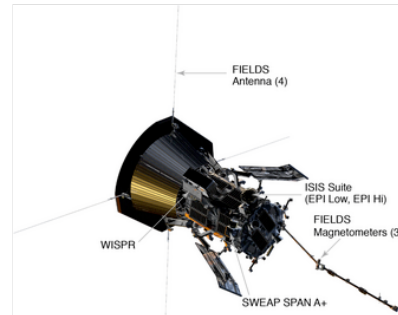


A visualization of PSP's projected orbit through 2029, assuming it continues to follow its current trajectory.

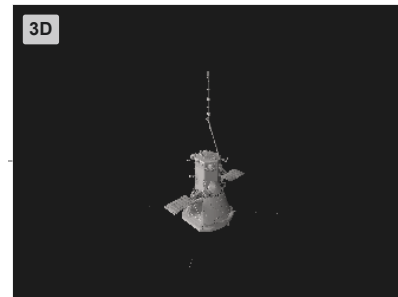
Instruments

Parker Solar Probe has four main instruments:^{[52][53]}

- **FIELDS** (Electromagnetic Fields Investigation). The instrument suite captures the scale and shape of electric and magnetic fields in the Sun's atmosphere. FIELDS measures waves and turbulence in the inner heliosphere with high time resolution to understand the fields associated with waves, shocks and magnetic reconnection, a process by which magnetic field lines explosively realign. FIELDS measures the electric field around the spacecraft with five antennas, four of which stick out beyond the spacecraft's heat shield and into the sunlight, where they experience temperatures of 1,370 °C (2,500 °F). The 2-meter-long (6 ft 7 in) antennas are made of a niobium alloy, which can withstand extreme temperatures. FIELDS measures electric fields across a broad frequency range both directly and remotely. Operating in two modes, the four sunlit antennas measure the properties of the fast and slow solar wind — the flow of solar particles constantly streaming out from the Sun. The fifth antenna, which sticks out perpendicular to the others in the shade of the heat shield, helps make a three-dimensional picture of the electric field at higher frequencies. The suite also has three magnetometers to assess the magnetic field. A search coil magnetometer, or SCM, measures how the magnetic field changes over time. Two identical fluxgate magnetometers, MAGi and MAGo, measure the large-scale coronal magnetic field. The fluxgate magnetometers are specialized for measuring the magnetic field further from the Sun where it varies at a slower rate, while the search coil magnetometer is necessary closer to the Sun where the field changes quickly, as it can sample the magnetic field at a rate of two million times per second. The Principal Investigator is Stuart Bale at the University of California, Berkeley.^[52]
- **IS $\odot$ IS** (Integrated Science Investigation of the Sun). The instrument uses two complementary instruments to measure particles across a wide range of energies. By measuring electrons, protons and ions, IS $\odot$ IS will understand the particles' lifecycles — where they came from, how they became accelerated and how they move out from the Sun through interplanetary space. The two energetic particle instruments on IS $\odot$ IS are called EPI-Lo and EPI-Hi (EPI stands for Energetic Particle Instrument). EPI-Lo measures the spectra of electrons and ions and identifies carbon, oxygen, neon, magnesium, silicon, iron and two isotopes of helium, He-3 and He-4. Distinguishing between helium isotopes will help determine which of several theorized mechanisms caused the particles' acceleration. The instrument has a design with an octagonal dome body supporting 80 viewfinders. Multiple viewfinders provide a wide field of view to observe low-energy particles. An ion that enters EPI-Lo through one of the viewfinders first passes through two carbon-polyimide-aluminum foils and then encounters a solid-state detector. Upon impact, the



Schematic view of all PSP's instruments



Parker Solar Probe 3D model

foils produce electrons, which are measured by a microchannel plate. Using the amount of energy left by the ion's impact on the detector and the time it takes the ions to pass through the sensor identifies the species of the particles. EPI-Hi uses three particle sensors composed of stacked layers of detectors to measure particles with energies higher than those measured by EPI-Lo. The front few layers are composed of ultra-thin silicon detectors made up of geometric segments, which allows for the determination of the particle's direction and helps reduce background noise. Charged particles are identified by measuring how deep they travel into the stack of detectors and how many electrons they pull off atoms in each detector, a process called ionization. At closest approach to the Sun, EPI-Hi will be able to detect up to 100,000 particles per second. The Principal Investigator is David McComas at Princeton University.^{[52][54]}

- **WISPR** (Wide-field Imager for Solar Probe). These optical telescopes acquire images of the corona and inner heliosphere. WISPR uses two cameras with radiation-hardened Active Pixel Sensor CMOS detectors. The camera's lenses are made of a radiation hard BK7, a common type of glass used for space telescopes, which is also sufficiently hardened against the impacts of dust. The Principal Investigator is Russell Howard at the Naval Research Laboratory.^[52]
- **SWEAP** (Solar Wind Electrons Alphas and Protons). This investigation will count the electrons, protons and helium ions, and measure their properties such as velocity, density, and temperature. Its main instruments are the Solar Probe Analyzers (SPAN, two electrostatic analyzers) and the Solar Probe Cup (SPC). SPC is a Faraday cup, a metal device that can catch charged particles in a vacuum. Peeking over the heat shield to measure how electrons and ions are moving, the cup is exposed to the full light, heat and energy of the Sun. The cup is composed of a series of highly transparent grids — one of which uses variable high voltages to sort the particles — above several collector plates, which measure the particles' properties. The variable voltage grid also helps sort out background noise, such as cosmic rays and photoionized electrons, which could otherwise bias the measurements. The grids, located near the front of the instrument, can reach temperatures of 1,650 °C (3,000 °F), glowing red while the instrument makes measurements. The instrument uses pieces of sapphire to electrically isolate different components within the cup. As it passes close to the Sun, SPC takes up to 146 measurements per second to accurately determine the velocity, density and temperature of the Sun's plasma. SPAN is composed of two instruments, SPAN-A and SPAN-B, which have wide fields of view to allow them to see the parts of space not observed by SPC. Particles encountering the detectors enter a maze that sends the particles through a series of deflectors and voltages to sort the particles based on their mass and charge. While SPAN-A has two components to measure both electrons and ions, SPAN-B looks only at electrons. The Principal Investigator is Justin Kasper at the University of Michigan and the Smithsonian Astrophysical Observatory.^[52]

An additional theoretical investigation named Heliospheric origins with Solar Probe Plus (HeliOSPP) starting from 2010 and ending in 2024 has the role of providing theoretical input and independent assessment of scientific performance to the Science Working Group (SWG) and the SPP Project to maximize the scientific return from the mission. The Principal Investigator is Marco Velli at the University of California, Los Angeles and the Jet Propulsion Laboratory; he also serves as the Observatory Scientist for the mission.^[38]

Mission

The Parker Solar Probe was launched on 12 August 2018, at 07:31 UTC. The spacecraft operated nominally after launching. During its first week in space it deployed its high-gain antenna, magnetometer boom, and electric field antennas.^[55] The spacecraft performed its first scheduled trajectory correction on 20 August 2018, while it was 8.8 million kilometers (5.5 million mi) from Earth, and travelling at 63,569 kilometers per hour (39,500 mph)^[56]

Instrument activation and testing began in early September 2018. On 9 September 2018, the two WISPR telescopic cameras performed a successful first-light test, transmitting wide-angle images of the background sky towards the Galactic Center.^[57]

The probe successfully performed the first of the seven planned Venus flybys on 3 October 2018, where it came within about 2,400 kilometers (1,500 mi) of Venus in order to reduce the probe's speed and orbit closer to the Sun.^[58]

Within each orbit of the Parker Solar Probe around the Sun, the portion within 0.25 AU is the Science Phase, in which the probe is actively and autonomously making observations. Communication with the probe is largely cut off in that phase.^{[59]:4} Science phases run for a few days both before and after each perihelion. They lasted 11.6 days for the earliest perihelion, and will drop to 9.6 days for the final, closest perihelion.^{[59]:8}

Much of the rest of each orbit is devoted to transmitting data from the science phase. But during this part of each orbit, there are still periods when communication is not possible. First, the requirement that the heat shield of the probe be pointed towards the Sun sometimes puts the heat shield between the antenna and Earth. Second, even when the probe is not particularly near the Sun, when the angle between the probe and the Sun, as seen from Earth, is too small, the Sun's radiation can overwhelm the communication link.^{[59]:11–14}

After the first Venus flyby, the probe was in an elliptical orbit with a period of 150 days (two-thirds the period of Venus), making three orbits while Venus makes two. After the second flyby, the period shortened to 130 days. After less than two orbits, only 198 days later, it encountered Venus a third time at a point earlier in the orbit of Venus. This encounter shortened its period to half of that of Venus, or about 112.5 days. After two orbits it met Venus a fourth time at about the same place, shortening its period to about 102 days.^[60]

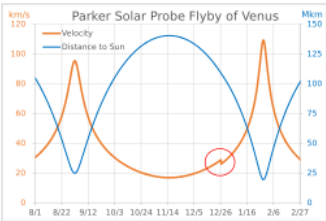
After 237 days, it met Venus for the fifth time and its period was shortened to about 96 days, three-sevenths that of Venus. It then made seven orbits while Venus made three. The sixth encounter, almost two years after the fifth, shortened its period down to 92 days, two-fifths that of Venus. After five more orbits (two orbits of Venus), it met Venus for the seventh and last time, decreasing its period to 88 or 89 days and allowing it to approach closer to the Sun.^[60]



Launch of the Parker Solar Probe in 2018

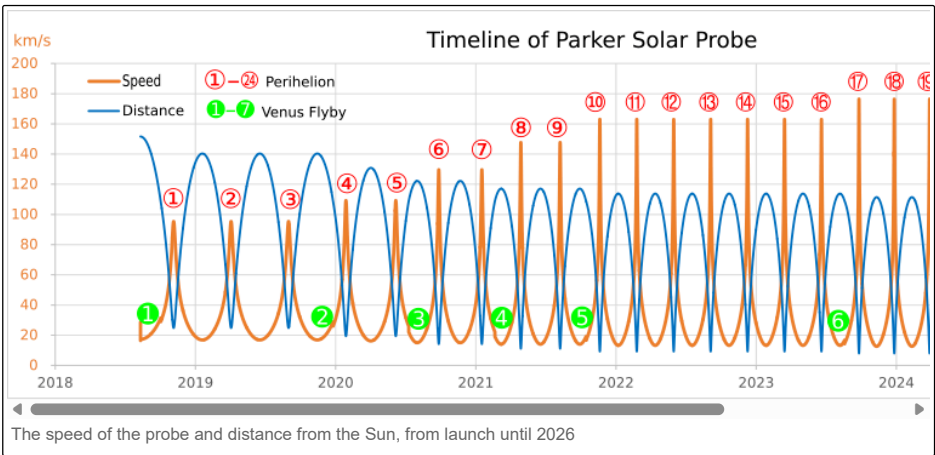


Artist's rendition of the Parker Solar Probe approaching the Sun



The second flyby of Venus on December 26, 2019. The velocity decreases by 2.9 km/s to 26 km/s (red circle), shifting the spacecraft to a new orbit closer to the Sun.

Timeline



The speed of the probe and distance from the Sun, from launch until 2026

List of events^{[60][38]:31}^[61]

Year	Date	Event	Perihelion distance (million km) ^[a]	Speed (km/s) ^[b]	Orbital period (days)	Notes
			Flyby altitude over Venus ^[c]	Leg of Parker's orbit ^[d]	Inside/Outside orbit of Venus ^[e]	
2018	12 August 07:31 UTC	Launch	151.6	–	174 ^[f]	
	3 October 08:44 UTC	Venus flyby #1	2548 km ^[g]	Inbound	Inside	Flybys 1 and 2 occur at the same point in Venus's orbit.
	6 November 03:27 UTC	Perihelion #1	24.8 × 10 ⁶ ^[h]	95	150	Solar encounter phase 31 October – 11 November ^[64]
2019	4 April 22:40 UTC	Perihelion #2				Solar encounter phase 30 March – 10 April ^[65]
	1 September 17:50 UTC ^[66]	Perihelion #3				Solar encounter phase 16 August – 20 September ^[i]
	26 December 18:14 UTC ^[68]	Venus flyby #2	3023 km	Inbound	Inside	Flybys 1 and 2 occur at the same point in Venus's orbit.
2020	29 January 09:37 UTC ^[69]	Perihelion #4	19.4 × 10 ⁶	109	130	Solar encounter phase 23 January – 29 February ^[70]
	7 June 08:23 UTC ^[71]	Perihelion #5				Solar encounter phase 9 May – 28 June ^[72]
	11 July 03:22 UTC ^[73]	Venus flyby #3	834 km	Outbound	Outside ^[j]	Flybys 3 and 4 occur at the same point in Venus's orbit.
	27 September	Perihelion #6	14.2 × 10 ⁶	129	112.5	
	17 January	Perihelion #7				
2021	20 February	Venus flyby #4	2392 km	Outbound	Outside	Flybys 3 and 4 occur at the same point in Venus's orbit.
	28 April	Perihelion #8	11.1 × 10 ⁶	147	102	First perihelion to enter the solar corona

	9 August	Perihelion #9				
	16 October	Venus flyby #5	3786 km	Inbound	Inside	Flybys 5 and 6 occur at the same point in Venus's orbit.
	21 November	Perihelion #10	9.2 × 10 ⁶	163	96	
2022	25 February	Perihelion #11				
	1 June	Perihelion #12				
	6 September	Perihelion #13				
	11 December	Perihelion #14				
2023	17 March	Perihelion #15	3939 km	Inbound	Inside	Flybys 5 and 6 occur at the same point in Venus's orbit.
	22 June	Perihelion #16				
	21 August	Venus flyby #6				
	27 September	Perihelion #17				
	29 December	Perihelion #18				
2024	30 March	Perihelion #19	7.9 × 10 ⁶	176	92	
	30 June	Perihelion #20				
	30 September	Perihelion #21				
	6 November	Venus flyby #7	317 km	Outbound	Outside	
	24 December	Perihelion #22	6.9 × 10 ⁶	192	88	
2025	22 March	Perihelion #23				
	19 June	Perihelion #24				
	15 September	Perihelion #25				
	12 December	Perihelion #26				

a. For altitude above the surface, subtract one solar radius ≈0.7 million km.